INCH-POUND

MIL-P-70505 (AR)
AMENDMENT 3
3 April 1989
SUPERSEDING
AMENDMENT 2
20 October 1987

MILITARY SPECIFICATION

PROJECTILE, 155MM: XM864
PROPELLANT GRAIN

This Amendment forms a part of Military Specification MIL-P-70505 (AR), dated 28 December 1984, and is approved for use within the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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* Add the designation "(M864)" in the title and change XM864 to "M864" wherever it appears in the specification.

* 2.1.1 Add the following military specification:

"MIL-A-70625 - Automated Acceptace Inspection Equipment for Non-Electronic Components, Design, Testing and Approval of"

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2.1.1 Add the following Military Standard:

"MIL-STD-453 Inspection, Radiographic".

* 2.1.1 Add the following drawings:

"INSPECTION EQUIPMENT DRAWINGS

9880422 Envelope Drawing Radiographic Equipment 155MM M864 Propellant Grain".

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* 2.2 Add the following new paragraphs:

"2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issued of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ANSI Z1.1 - Guide for Quality Control Charts
ANSI Z1.1 - Control Chart Method of Analyzing Data
ANSI Z1.1 - Control Chart Method fo Controlling Quality During Production

(Applications for copies should be addressed to American National Standards Institute, 1430 Broadway, N.Y., N.Y. 10018)".

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3.3 Delete "930 ± 35 cal/g" and substitute "950 ± 50 cal/g".

3.4 Delete in its entirety and substitute the following:

"3.4 Strand burning rate. Strand burning rate tests shall be preformed as shown in TABLE I. Only results from the 70°F, 14.7 psia test shall be used for lot acceptance.

TABLE I. Strand burning rate.

Temperature (± 5°F)	Pressure (psia ± 10%)	Rate (mm/sec)
-50	14.7	INFO
70	14.7	1.00 ± .10
145	14.7	INFO

* 3.6.2.1 Add the following new paragraph:

"3.6.2.1 Inhibitor flash. There shall be no evidence of inhibitor flash as indicated on the applicable drawing."

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* 3.6.3 Delete in its entirety and substitute the following:

"3.6.3 Internal defects. Each grain segment (individual split) shall contain no cracks, voids or fissures measuring more than .20 inch in any direction and no foreign material measuring more than 0.20 inch in any direction. The total combined length of all cracks, voids and fissures in any segment shall not exceed .30 inch and the combined length of all foreign material (maximum dimension each particle) shall not exceed 0.20 inch."

3.7 Delete paragraphs a and b and substitute the following:

"a. Modulus of elasticity of 700 psi minimum per sample average.

b. Ultimate stress of 170 psi minimum per sample average."

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4.3.1 Change "TABLE I" to TABLE II".

4.3.1 Table I: Change strand burning rate samples from "12" to "18". Add the following in the paragraph reference section of the strand burning test:

"(14.7 psia tests only, 6 at each temperature)".

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* 4.4.1.1 Delete in its entirety and substitute the following:

"4.4.1.1 Propellant or inhibitor batch. A batch of material, either inhibitor or propellant, is defined as a homogeneous quantity of raw materials that are blended and processed in at one time. As a maximum, a batch shall consist of materials blended at one time in one mixing or blending bowl. A batch shall contain individual ingredients from not more than one lot from one manufacturer for each ingredient. After a batch has been accepted following physical and mechanical properties testing, it may be combined with other accepted batches provided that the batches are homogeneous."

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* 4.4.2.c Add the following paragraph:

"Statistical process control (SPC). For those dimensions identified in the Classification of Defects Tables by the reference, "(see Appendix A)", the contractor must utilize the principals of SPC as defined herein, ANSI Z1.1, ANSI Z1.2, ANSI Z1.3, and in Appendix A. When the contractor satisfies the control conditions identified in Appendix A acceptance will then be by SPC. The SPC sample size and frequency of selection shall be determined by the contractor. If the SPC sample is to be less than 4 items per hour, then the approval of the procuring agency must be sought prior to the elimination of sampling inspection for acceptance. The contractor may request to implement SPC acceptance for any other feature that is not annotated in the classification for defects by requesting approval of the procurement agency and by meeting all the conditions specified in this specification. When not accepting product by SPC, acceptance of product for those features identified with references to Appendix A, will only be in accordance with MIL-STD-414 or MIL-STD-1235 plans CSP-1, -2, -T using variable measurement capable gages."

* 4.4.2.1 Examination or test. Add "(see Appendix A)" after major defects 101 and 102.

4.4.2.3 Change title to "precast Inhibitor, Piror to assembly to grain". Delete special defect A and substitute the following:

"A. Inhibitor prior to bonding to propellant (applies to pre-cast inhibitor only) 100%, 3.6.2, Visual".

* 4.4.2.4 Add the following defect:

"Special c	Inhibitor Flash Present	100%	3.6.2.1	Visual"

Major 104: Change "four" to "two".

Major 106: Change "4" to "2".

Major 107: Change "8" to "4".

Minor 201: Change "Location of spacers" to "True position of spacers".

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4.4.2.5 Change Minor 202 from "100%" to ".65%".

4.5.1 Delete in its entirety and substitute the following:

"4.5.1 Strand burning characteristics of propellant. Along with propellant grains cast from each batch of mix of chemicals, nine test strands shall be taken from the same propellant batch and cured in the same chamber under the same conditions for the same length of time. The strands shall be tested in accordance with method T803.1 of MIL-STD-286 or an approved alternate (see 6.5). Three strands shall be tested at each of the temperature - pressure combinations in 3.4. If the average of three strands fails to meet the requirements of 3.4, the batch shall be rejected."

* 4.5.5 Delete "3.6.2" in the last sentence and substitute "3.6". Add the following sentence at the end of the paragraph:

"Equipment and procedures used for radiographic inspection shall be in accordance with MIL-STD-453. Grain segments may be radiographed prior to assembly of spacers."

* 4.5.5.1 Add the following new paragraph:

"4.5.5.1 Automated grain inspection. Every propellant grain shall be radiographically inspected for defects by using approved hard x-ray procedures. Strands and test methods for the automated inspection system shall be in accordance with drawing 9880422, and MIL-A-70625. These hard x-rays will then be used to rate the automated grain inspection system by comparing the hard x-ray side-by-side to the automated system's detection capability of the very same grain. This comparison of hard x-ray to the digital display of the automated grain inspection system will be conducted and recorded for information purposes only. All regular production accept/reject decisions will be made with the use of the hard x-ray only. When enough information is gathered to prove the reliability of the automated grain inspection system is equal to or greater than hard x-ray, it will be the option of the cognizant engineering facility to allow continued use of hard x-ray procedures or to change over to full time use of automated grain inspection."

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* 4.5.5.2 Add the following new paragraph:

"4.5.5.2 Calibration of radiograph equipment. The automated grain inspection system shall be calibrated by using the standard described in drawing 9880422 at the start and finish of eah production shift, or change in operator, as well as at one (1) hour intervals during production. If the inspection process is determined to be inadequate, the condition shall then be corrected and all items inspected since the last acceptable calibration check shall be reinspected."

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4.5.10.3 Change paragraph number to "4.5.6.3".

4.5.10.4 Change paragraph number to "4.5.6.4".

6.4 Change "SMCAR-LCU-SI" to "SMCAR-FSA-IM".

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* Add new Appendix A included herein, pages (18) through (20).

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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APPENDIX A

10. SCOPE

10.1 Scope. This appendix details the procedure for implementation of statistical process control for articles covered in this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 Application of statistical process control. The contractor will for the identified dimensions maintain Xbar and R SPC charts. After the contractor has demonstrated that his process for manufacturing a particular feature is in control, as defined in paragraph A.1.1; he will discontinue acceptance sampling inspection in favor of acceptance by SPC. However, SPC charting will be maintained for all identified dimensions whether or not the process is in control. It is expected that the contractor will make every effort to bring each process into control. It is required that the contractor re-evaluate his established control limits on a monthly basis and make those changes warranted by the data. At no time should his process parameters vary such that it will fail to meet the requirements established in subsequent paragraphs. Should this happen, the contractor will return to acceptance by sampling.

30.1.1 Implementation of SPC for acceptance of product. For those dimensions identified in the classification of defects tables as being applicable for elimination of lot sampling acceptance, the contractor shall implement, on a dimension by dimension basis, acceptance by SPC after he conforms to each of the following steps:

a. The contractor performs an indepth process capability study to establish the mean process average (Xdouble bar), Rbar, upper and lower control limits. This study will consist of at least 40 subgroups of data collection.

b. The contractor demonstrates that his process is under control for at least 20 consecutive production days. Control is defined as remaining within the established control parameters and not exhibiting any of the out of control conditions defined in paragraph A.1.2.a through A.1.2.e more than two times on an individual character basis.

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c. Thirty days consecutive production have passed acceptance sampling for the feature.

d. The process has a demonstrated Cpk equal to or greater than 1.3 as demonstrated when conducting the above test. Cpk is defined as follows:

$$Cpk = \frac{\text{process mean - nearest specification limit}}{3 \text{ sigma}}$$

sigma = process standard deviation (calculated by the range method per the referenced ANSI documents)

e. The contractor documents his detailed process description. This process description must be submitted to Commander, AMCCOM, Attn: AMSMC-QAR-Q (D), Picatinny Arsenal, New Jersey 07806-5000 for review and disapproval if the description does not meet its objectives. That is the process description will identify all known contributing variables and their respective controls that are necessary to maintain the specific dimension (i.e., weight, temperature, machine speed, time, tool life, solution proportion, etc.). The government will furnish the contractor with written notice of the acceptability of his process controls. A notice of control acceptability is required prior to the implementation of acceptance by SPC.

30.1.2 Process control analysis. Once SPC acceptance is implemented, the contractor will evaluate his process for evidence of nonrandom variation (e.g. trends, extreme values, etc.). Should any out of control condition be detected, the product produced from the last sample till the process is brought back into control will be isolated for analysis. The contractor will bring the process back into control or shut the process down. Upon determining the cause of the out of control condition, the contractor will determine if the product manufactured and isolated could be out of engineering tolerance. If the out of control condition did not result in the production of nonconforming material, the material will be released. If the out of control condition could have created nonconforming product, the contractor will sample the product using MIL-STD-105 and the AQL specified in the classification of defects table to determine the acceptability of the isolated product. This evaluation of out of control product requires the review and approval of the local government quality assurance representative. As a minimum, the contractor will look for any of the following out of control indications:

a. If any point goes outside the Xbar or R chart control limits.

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b. If 9 points in a row are either continually above or continually below the X double bar.

c. If 6 points in a row are either continually increasing or continually decreasing on the Xbar chart.

d. If 6 points in a row are increasing on the range chart.

e. If any individual sample, Xi, falls outside of the specification limits.

30.1.2.1 Process capability review. At least once a month the contractor will review his process data and establish updated control limits.

30.1.3 Mandatory return to sampling acceptance. Acceptance by SPC will be discontinued in accordance with the following paragraphs. In any case, control charting will be continued. When this occurs, the contractor may reintroduce acceptance by SPC by repeating the steps outlined in paragraph A.1.1.

30.1.3.1 Out of control incidents. If for any individual process, an out of control incident is identified 3 times within 5 consecutive shifts or 5 times within 30 consecutive shifts; and the analysis indicated that nonconforming product may have been manufactured, the contractor will return to acceptance by sampling for the pertinent feature.

30.1.3.2 Process interruption. If the process is interrupted for more than 10 calendar days, the contractor will return to acceptance by sampling for that pertinent feature.

30.1.3.3 Loss of control. If either by testing or data review it is determined that the Cpk is no longer at the required minimum of 1.3, the contractor will return to acceptance by sampling for the pertinent feature.

30.1.4 Records. The contractor will maintain all records dealing with SPC in his files for a minimum of 5 years after completion of the contract.

30.2 Government auditing. In order to verify the state of control of the process, it is recommended that government representatives select samples at unscheduled times. The sample mean and range should then be compared to the data being recorded by the contractor. If the data is not consistent with the reported data, corrective action should be sought and if necessary, screening of material should be requried.

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